

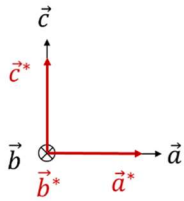
Q1) Interplanar spacing and angle in 3D Bravais lattices: (50 pts)

- a) Starting from the definition of the reciprocal space lattice vector, derive the interplanar distance d_{hkl} equation with respect to (hkl) planes in the cubic system with lattice constant a .

Solution:

Step 1: Compute $a^*, b^*, c^*, \alpha^*, \beta^*, \gamma^*$ in terms of $a, b, c, \alpha, \beta, \gamma$

Cubic system: $\alpha=\alpha^*=\beta=\beta^*=\gamma=\gamma^*=90^\circ$ & $a=b=c$

$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{V} = \frac{\vec{b} \times \vec{c}}{\vec{a} \cdot (\vec{b} \times \vec{c})}$$


$$|\vec{a}| = |\vec{b}| = |\vec{c}| \quad \& \quad |\vec{a}^*| = |\vec{b}^*| = |\vec{c}^*|$$

$$\vec{a}^* = \frac{a\vec{y} \times a\vec{z}}{a^3} = \frac{1}{a}\vec{x} \quad |\vec{a}^*| = \sqrt{\vec{a}^* \cdot \vec{a}^*} = \sqrt{\frac{\vec{x} \cdot \vec{x}}{a^2}} = \frac{1}{a}$$

Step 2: Compute the reciprocal metric tensor for the cube

$$\Gamma^* = \begin{bmatrix} a^{*2} & a^*b^*\cos\gamma^* & a^*c^*\cos\beta^* \\ a^*b^*\cos\gamma^* & b^{*2} & b^*c^*\cos\alpha^* \\ a^*c^*\cos\beta^* & b^*c^*\cos\alpha^* & c^{*2} \end{bmatrix}$$

$$\Gamma_{cubic}^* = \begin{bmatrix} 1/a^2 & 0 & 0 \\ 0 & 1/a^2 & 0 \\ 0 & 0 & 1/a^2 \end{bmatrix}$$

Step 3: Compute the magnitude of the reciprocal vector $|\mathbf{k}_{hkl}| = 1/d_{hkl}$

$$|\vec{k}_{hkl}^*| = \frac{1}{d_{hkl}} \quad \vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l][\Gamma^*] \begin{bmatrix} h \\ k \\ l \end{bmatrix}$$

$$\frac{1}{d_{hkl}^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

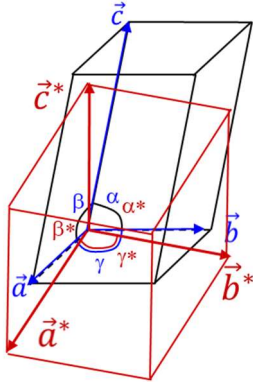
Step 4: Solve for d_{hkl}

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

b) Starting from the definition of the reciprocal space lattice vector, express the interplanar distance d_{hkl} between (hkl) planes in the triclinic system.

Solution: For triclinic, it's a bit more complex but the same process.

Step 1: Compute a^* , b^* , c^* , α^* , β^* , γ^* in terms of a , b , c , α , β , γ



$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{V} = \frac{\vec{b} \times \vec{c}}{\vec{a} \cdot (\vec{b} \times \vec{c})} \quad \vec{b}^* = \frac{\vec{c} \times \vec{a}}{V} = \frac{\vec{c} \times \vec{a}}{\vec{a} \cdot (\vec{b} \times \vec{c})} \quad \vec{c}^* = \frac{\vec{a} \times \vec{b}}{V} = \frac{\vec{a} \times \vec{b}}{\vec{a} \cdot (\vec{b} \times \vec{c})}$$

$$V = abc (1 - \cos^2 \alpha - \cos^2 \beta - \cos^2 \gamma + 2 \cos \alpha \cos \beta \cos \gamma)^{1/2}$$

$$\cos \alpha^* = \frac{\cos \beta \cos \gamma - \cos \alpha}{\sin \beta \sin \gamma} \quad \cos \beta^* = \frac{\cos \alpha \cos \gamma - \cos \beta}{\sin \alpha \sin \gamma} \quad \cos \gamma^* = \frac{\cos \alpha \cos \beta - \cos \gamma}{\sin \alpha \sin \beta}$$

Step 2: Compute the reciprocal metric tensor for a triclinic crystal

$$\Gamma^* = \begin{bmatrix} a^{*2} & a^* b^* \cos \gamma^* & a^* c^* \cos \beta^* \\ a^* b^* \cos \gamma^* & b^{*2} & b^* c^* \cos \alpha^* \\ a^* c^* \cos \beta^* & b^* c^* \cos \alpha^* & c^{*2} \end{bmatrix}$$

Step 3: Compute the magnitude of the reciprocal vector $|\mathbf{k}_{hkl}| = 1/d_{hkl}$

$$\frac{abc}{V^2} \begin{pmatrix} bcsin^2 \alpha / a & c(\cos \alpha \cos \beta - \cos \gamma) & b(\cos \alpha \cos \gamma - \cos \beta) \\ c(\cos \alpha \cos \beta - \cos \gamma) & acsin^2 \beta / b & a(\cos \beta \cos \gamma - \cos \alpha) \\ b(\cos \alpha \cos \gamma - \cos \beta) & a(\cos \beta \cos \gamma - \cos \alpha) & absin^2 \gamma / c \end{pmatrix}$$

Step 4: Solve for d_{hkl}

$$\begin{aligned} \frac{1}{d_{hkl}^2} &= h^2 \frac{b^2 c^2 \sin^2 \alpha}{V^2} + k^2 \frac{a^2 c^2 \sin^2 \beta}{V^2} + l^2 \frac{a^2 b^2 \sin^2 \gamma}{V^2} \\ &+ 2hk \frac{abc^2 (\cos \alpha \cos \beta - \cos \gamma)}{V^2} + 2kl \frac{a^2 bc (\cos \beta \cos \gamma - \cos \alpha)}{V^2} \\ &+ 2lh \frac{ab^2 c (\cos \gamma \cos \alpha - \cos \beta)}{V^2} \end{aligned}$$

- c) Compute the angle between the (321) and (112) planes in
a. {1, 1, 1, 90, 90, 90}

Solution:

$$\cos \theta = \frac{\vec{k}_{h_1 k_1 l_1}^* \cdot \vec{k}_{h_2 k_2 l_2}^*}{|\vec{k}_{h_1 k_1 l_1}^*| |\vec{k}_{h_2 k_2 l_2}^*|}$$

$$\vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l][\Gamma^*] \begin{bmatrix} h \\ k \\ l \end{bmatrix} \quad \vec{k}_{h_1 k_1 l_1}^* \cdot \vec{k}_{h_2 k_2 l_2}^* = [h_1 \quad h_1 \quad l_1][\Gamma^*] \begin{bmatrix} h_2 \\ k_2 \\ l_2 \end{bmatrix}$$

$$\Gamma_{cubic}^* = \begin{bmatrix} 1/a^2 & 0 & 0 \\ 0 & 1/a^2 & 0 \\ 0 & 0 & 1/a^2 \end{bmatrix}$$

For a cubic system:

As seen in class...

$$\cos \theta = \frac{h_1 h_2 + k_1 k_2 + l_1 l_2}{\sqrt{h_1^2 + k_1^2 + l_1^2} \sqrt{h_2^2 + k_2^2 + l_2^2}}$$

$$\cos \theta = \frac{(3 \cdot 1) + (2 \cdot 1) + (1 \cdot 2)}{\sqrt{3^2 + 2^2 + 1^2} \sqrt{1^2 + 1^2 + 2^2}} = \frac{7}{\sqrt{14}\sqrt{6}}$$

$$\theta = 40.2^\circ$$

- b. {1, 1, 2, 90, 90, 90}

Solution:

$$\cos \theta = \frac{\vec{k}_{h_1 k_1 l_1}^* \cdot \vec{k}_{h_2 k_2 l_2}^*}{|\vec{k}_{h_1 k_1 l_1}^*| |\vec{k}_{h_2 k_2 l_2}^*|} \quad \vec{k}_{h_1 k_1 l_1}^* \cdot \vec{k}_{h_2 k_2 l_2}^* = [h_1 \quad k_1 \quad l_1][\Gamma^*] \begin{bmatrix} h_2 \\ k_2 \\ l_2 \end{bmatrix}$$

For a tetragonal system, we can show that

$$\Gamma_{tetragonal}^* = \begin{bmatrix} a^{*2} & 0 & 0 \\ 0 & a^{*2} & 0 \\ 0 & 0 & c^{*2} \end{bmatrix}$$

$$\vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l][\Gamma^*] \begin{bmatrix} h \\ k \\ l \end{bmatrix} \quad \& \quad \boxed{|\vec{k}_{hkl}^*| = \frac{1}{d_{hkl}}}$$

Let's compute $|\vec{k}_{hkl}^*|$:

$$\begin{aligned} |\vec{k}_{hkl}^*|^2 &= \vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l] \begin{bmatrix} \frac{1}{a^2} & 0 & 0 \\ 0 & \frac{1}{a^2} & 0 \\ 0 & 0 & \frac{1}{c^2} \end{bmatrix} \begin{bmatrix} h \\ k \\ l \end{bmatrix} = [h \quad k \quad l] \begin{bmatrix} \frac{h}{a^2} \\ \frac{k}{a^2} \\ \frac{l}{c^2} \end{bmatrix} \\ &= \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2} \\ d_{hkl} &= \frac{ac}{\sqrt{c^2(h^2 + k^2) + a^2 l^2}} \end{aligned}$$

As well as

$$\begin{aligned} \vec{k}_{h_1 k_1 l_1}^* \cdot \vec{k}_{h_2 k_2 l_2}^* &= [h_1 \quad k_1 \quad l_1][\Gamma^*] \begin{bmatrix} h_2 \\ k_2 \\ l_2 \end{bmatrix} \\ &= [h_1 \quad k_1 \quad l_1] \begin{bmatrix} \frac{1}{a^2} & 0 & 0 \\ 0 & \frac{1}{a^2} & 0 \\ 0 & 0 & \frac{1}{c^2} \end{bmatrix} \begin{bmatrix} h_2 \\ k_2 \\ l_2 \end{bmatrix} = [h_1 \quad k_1 \quad l_1] \begin{bmatrix} \frac{h_2}{a^2} \\ \frac{k_2}{a^2} \\ \frac{l_2}{c^2} \end{bmatrix} \\ &= \frac{h_1 h_2 + k_1 k_2}{a^2} + \frac{l_1 l_2}{c^2} \end{aligned}$$

$$\vec{k}_{h_1k_1l_1}^* \cdot \vec{k}_{h_2k_2l_2}^* = \frac{h_1h_2 + k_1k_2}{a^2} + \frac{l_1l_2}{c^2} = \frac{(h_1h_2 + k_1k_2)c^2 + l_1l_2a^2}{(ac)^2}$$

$$\cos \theta = \frac{\vec{k}_{h_1k_1l_1}^* \cdot \vec{k}_{h_2k_2l_2}^*}{|\vec{k}_{h_1k_1l_1}^*| |\vec{k}_{h_2k_2l_2}^*|}$$

$$\begin{aligned} \cos \theta &= \frac{\frac{(h_1h_2 + k_1k_2)c^2 + l_1l_2a^2}{(ac)^2}}{\frac{\sqrt{c^2(h_1^2 + k_1^2) + a^2l_1^2}}{ac} \frac{\sqrt{c^2(h_2^2 + k_2^2) + a^2l_2^2}}{ac}} \\ &= \frac{(h_1h_2 + k_1k_2)c^2 + l_1l_2a^2}{\sqrt{c^2(h_1^2 + k_1^2) + a^2l_1^2} \sqrt{c^2(h_2^2 + k_2^2) + a^2l_2^2}} \end{aligned}$$

{1, 1, 2, 90, 90, 90}

$$\begin{aligned} \cos \theta &= \frac{(3 \cdot 1 + 2 \cdot 1)(2)^2 + (1 \cdot 2)(1)^2}{\sqrt{2^2(3^2 + 2^2) + 1^21^2} \sqrt{2^2(1^2 + 1^2) + 1^22^2}} = \frac{20 + 2}{\sqrt{4(13) + 1} \sqrt{4(2) + 4}} = \frac{22}{\sqrt{53}\sqrt{12}} \\ &= 0.872 \end{aligned}$$

$$\theta = 29.3^\circ$$

c. {1, 1, 2, 90, 75, 85}

Solution:

$$\cos \theta = \frac{\vec{k}_{h_1k_1l_1}^* \cdot \vec{k}_{h_2k_2l_2}^*}{|\vec{k}_{h_1k_1l_1}^*| |\vec{k}_{h_2k_2l_2}^*|}$$

$$\vec{k}_{hkl}^* \cdot \vec{k}_{hkl}^* = \frac{1}{d_{hkl}^2} = [h \quad k \quad l][\Gamma^*] \begin{bmatrix} h \\ k \\ l \end{bmatrix} \quad \Gamma^* = \begin{bmatrix} a^{*2} & a^*b^*\cos\gamma^* & a^*c^*\cos\beta^* \\ a^*b^*\cos\gamma^* & b^{*2} & b^*c^*\cos\alpha^* \\ a^*c^*\cos\beta^* & b^*c^*\cos\alpha^* & c^{*2} \end{bmatrix}$$

We can work from our expression we got in part b.

$$|\vec{k}_{hkl}^*| = \frac{1}{d_{hkl}}$$

$$\frac{1}{d_{hkl}^2} = h^2 \frac{b^2 c^2 \sin^2 \alpha}{V^2} + k^2 \frac{a^2 c^2 \sin^2 \beta}{V^2} + l^2 \frac{a^2 b^2 \sin^2 \gamma}{V^2} \\ + 2hk \frac{abc^2 (\cos \alpha \cos \beta - \cos \gamma)}{V^2} + 2kl \frac{a^2 bc (\cos \beta \cos \gamma - \cos \alpha)}{V^2} \\ + 2lh \frac{ab^2 c (\cos \gamma \cos \alpha - \cos \beta)}{V^2}$$

$$\cos \theta = \frac{\vec{k}_{h_1 k_1 l_1}^* \cdot \vec{k}_{h_2 k_2 l_2}^*}{\left| \vec{k}_{h_1 k_1 l_1}^* \right| \left| \vec{k}_{h_2 k_2 l_2}^* \right|}$$

$$= \frac{1}{\left| \vec{k}_{h_1 k_1 l_1}^* \right| \left| \vec{k}_{h_2 k_2 l_2}^* \right|} [h_1 h_2 b^2 c^2 \sin^2 \alpha + k_1 k_2 a^2 c^2 \sin^2 \beta + l_1 l_2 a^2 b^2 \sin^2 \gamma \\ + abc^2 (h_1 k_2 + h_2 k_1) (\cos \alpha \cos \beta - \cos \gamma) \\ + ab^2 c (h_1 l_2 + h_2 l_1) (\cos \alpha \cos \gamma - \cos \beta) \\ + a^2 bc (k_1 l_2 + k_2 l_1) (\cos \beta \cos \gamma - \cos \alpha)]$$

{1, 1, 2, 90, 75, 85}, between the (321) and (112) planes

$$\cos \theta = \frac{1}{\left| \vec{k}_{h_1 k_1 l_1}^* \right| \left| \vec{k}_{h_2 k_2 l_2}^* \right|} [12 + 8 \sin^2 75 + 2 \sin^2 85 + 4(3 + 2)(-\cos 85) \\ + 2(6 + 1)(-\cos 75) + 2(4 + 1)(\cos 75 \cos 85)] = \frac{16.31}{\sqrt{44.81} \sqrt{9.11}} \\ = 0.8072$$

$$\theta = 36.18^\circ$$

Q2) Identify family of planes: (30 pts)

What is the family of planes {hkl} corresponding with an interplanar spacing of $d = 1.246 \text{ \AA}$ in nickel (Ni) with $a = 3.524 \text{ \AA}$? Hint: the solution requires identifying several (hkl) planes that fulfill the interplanar spacing and concluding the family of planes they belong to.

Solution:

Ni is a cube, hence the d_{hkl} relationship to hkl indices is known (below). We need to compute a table with trial Miller indices and compute d_{hkl}

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Trial index	$h^2 + k^2 + l^2$	d_{hkl} (Å)	Equal to 1.246 Å?
100	1	3.524	N
110	2	2.492	N
111	3	2.035	N
200	4	1.762	N
210	5	1.576	N
211	6	1.434	N
220	8	1.246	Y

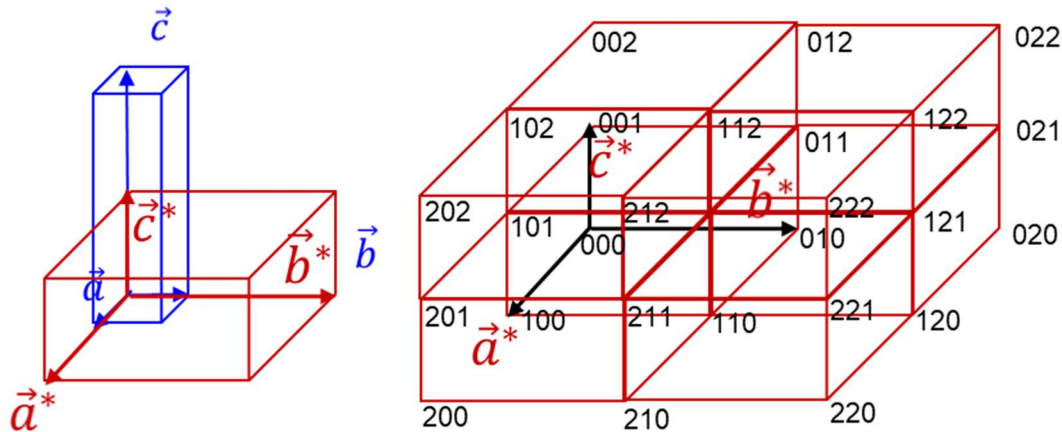
220, 202, 022, etc. will yield the same d-spacing, since $a=b=c$. Consequently, the family of planes in question is {220}.

Q3) Reciprocal space lattice: (30 pts)

Starting with a tetragonal P lattice, with lattice parameters a and c , **calculate** the reciprocal lattice vectors, **draw** the reciprocal Bravais lattice, and **indicate** the lattice vectors $\mathbf{a}^*$, $\mathbf{b}^*$ and $\mathbf{c}^*$. **Draw** a 3D lattice corresponding to $2 \times 2 \times 2$ unit cells (8 unit cells in total). Place the origin at the center and using the reciprocal lattice vector and the addition rule, **index** all reciprocal lattice points.

Solution:

We can show, following in-class exercises 1 & 2, that $a^* = 1/a$, $b^* = 1/a$ and $c^* = 1/c$. The vectors are also parallel such that $\mathbf{a}^* \parallel \mathbf{a}$, $\mathbf{b}^* \parallel \mathbf{b}$, and $\mathbf{c}^* \parallel \mathbf{c}$.



Q4) Reciprocal metric tensor and x-ray diffraction: (60 pts)

In 2004, NASA's Mars rover Spirit examined the Columbia Hills region of Mars and found the mineral goethite. Goethite is known to form only in the presence of water and the discovery was taken as evidence for water activity on Mars. Its chemical composition is iron(III) oxide-hydroxide, its space group is Pbnm, and the values of its lattice constants are $a = 0.4596$, $b = 0.9957$, and $c = 0.3021$ nm. Determine the following:



(a) The reciprocal lattice parameters of goethite.

Solution:

Compute a^* , b^* , c^* , α^* , β^* , γ^* in terms of a , b , c , α , β , γ

Orthorhombic system: $\alpha = \alpha^* = \beta = \beta^* = \gamma = \gamma^* = 90^\circ$ & $a \neq b \neq c$

$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{V} = \frac{\vec{b} \times \vec{c}}{\vec{a} \cdot (\vec{b} \times \vec{c})} \quad \vec{b}^* = \frac{\vec{c} \times \vec{a}}{V} = \frac{\vec{c} \times \vec{a}}{\vec{a} \cdot (\vec{b} \times \vec{c})} \quad \vec{c}^* = \frac{\vec{a} \times \vec{b}}{V} = \frac{\vec{a} \times \vec{b}}{\vec{a} \cdot (\vec{b} \times \vec{c})}$$

$$V = abc (1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2 \cos \alpha \cos \beta \cos \gamma)^{1/2}$$

(b) Draw a 3D lattice corresponding to $2 \times 2 \times 2$ unit cells, placing the origin at the center and index all reciprocal lattice points.

Solution:

(e) The specific reciprocal metric tensor for goethite.

Solution:

$$\Gamma_{orthor.}^* = \begin{bmatrix} \frac{1}{a^2} & 0 & 0 \\ 0 & \frac{1}{b^2} & 0 \\ 0 & 0 & \frac{1}{c^2} \end{bmatrix} = \begin{bmatrix} 4.73 \text{ nm}^{-2} & 0 & 0 \\ 0 & 1.01 \text{ nm}^{-2} & 0 \\ 0 & 0 & 10.96 \text{ nm}^{-2} \end{bmatrix}$$

(f) The values for the interplanar spacing between (020), (200), and (111) planes in goethite.

Solution: Two approaches, use d_{hkl} equation or matrix approach.

$$d_{020} = \frac{0.4596 \times 0.9957 \times 0.3021 \text{ nm}^3}{\sqrt{0 + 4(0.4596 \text{ nm} \times 0.3021 \text{ nm})^2 + 0}} = \frac{0.138 \text{ nm}^3}{0.278 \text{ nm}^2} = 0.496 \text{ nm}$$

$$d_{200} = \frac{0.4596 \times 0.9957 \times 0.3021 \text{ nm}^3}{\sqrt{4(0.9957 \text{ nm} \times 0.3021 \text{ nm})^2 + 0 + 0}} = \frac{0.138 \text{ nm}^3}{0.602 \text{ nm}^2} = 0.229 \text{ nm}$$

d_{111}

$$\begin{aligned} &= \frac{0.4596 \times 0.9957 \times 0.3021 \text{ nm}^3}{\sqrt{(0.9957 \text{ nm} \times 0.3021 \text{ nm})^2 + (0.4596 \text{ nm} \times 0.3021 \text{ nm})^2 + (0.4596 \text{ nm} \times 0.9957 \text{ nm})^2}} \\ &= \frac{0.138 \text{ nm}^3}{0.565 \text{ nm}^2} = 0.244 \text{ nm} \end{aligned}$$

(g) The angle between the plane normal (101) and $(\bar{1}01)$ in goethite.

Solution:

$$\cos \theta = \frac{\vec{k}_{h1k1l1}^* \cdot \vec{k}_{h2k2l2}^*}{|\vec{k}_{h1k1l1}^*| |\vec{k}_{h2k2l2}^*|}$$

$$|\vec{k}_{hkl}|^2 = \frac{1}{d_{hkl}^2} = [h \quad k \quad l] \begin{bmatrix} \frac{1}{a^2} & 0 & 0 \\ 0 & \frac{1}{b^2} & 0 \\ 0 & 0 & \frac{1}{c^2} \end{bmatrix} \begin{bmatrix} h \\ k \\ l \end{bmatrix} = [h \quad k \quad l] \begin{bmatrix} \frac{h}{a^2} \\ \frac{k}{b^2} \\ \frac{l}{c^2} \end{bmatrix} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

$$\begin{aligned}
\vec{k}_{h_1k_1l_1} \cdot \vec{k}_{h_2k_2l_2} &= [h_1 \quad k_1 \quad l_1] \begin{bmatrix} \frac{1}{a^2} & 0 & 0 \\ 0 & \frac{1}{b^2} & 0 \\ 0 & 0 & \frac{1}{c^2} \end{bmatrix} \begin{bmatrix} h_2 \\ k_2 \\ l_2 \end{bmatrix} = [h_1 \quad k_1 \quad l_1] \begin{bmatrix} \frac{h_2}{a^2} \\ \frac{k_2}{b^2} \\ \frac{l_2}{c^2} \end{bmatrix} \\
&= \frac{h_1 h_2}{a^2} + \frac{k_1 k_2}{b^2} + \frac{l_1 l_2}{c^2} \\
\cos \theta &= \frac{\frac{h_1 h_2}{a^2} + \frac{k_1 k_2}{b^2} + \frac{l_1 l_2}{c^2}}{\sqrt{\frac{h_1^2}{a^2} + \frac{k_1^2}{b^2} + \frac{l_1^2}{c^2}} \sqrt{\frac{h_2^2}{a^2} + \frac{k_2^2}{b^2} + \frac{l_2^2}{c^2}}} \\
&= \frac{h_1 h_2 (bc)^2 + k_1 k_2 (ac)^2 + l_1 l_2 (ab)^2}{\sqrt{h_1^2 (bc)^2 + k_1^2 (ac)^2 + l_1^2 (ab)^2} \sqrt{h_2^2 (bc)^2 + k_2^2 (ac)^2 + l_2^2 (ab)^2}} \\
&= \frac{1(-1)(0.9957 \times 0.3021)^2 + 0 + 1(1)(0.4596 \times 0.9957)^2}{\sqrt{1(0.9957 \times 0.3021)^2 + 0 + 1(0.4596 \times 0.9957)^2} \sqrt{1(0.9957 \times 0.3021)^2 + 0 + 1(0.4596 \times 0.9957)^2}} \\
&= \frac{0.1189}{0.5476 \times 0.5476} = 0.3965 \\
\theta &= 66.64
\end{aligned}$$

(h) The diffraction pattern of a goethite crystal is measured in a Bragg-Brentano geometry using a Cu K α ($\lambda=1.245$ Å). Prepare a table of Miller indices of planes, interplanar spacings and diffraction angles for $0 < 2\theta < 60^\circ$.

Bragg's law: $2d_{hkl} \sin \theta_{hkl} = \lambda$; $d_{hkl} = \frac{abc}{\sqrt{h^2(bc)^2 + k^2(ac)^2 + l^2(ab)^2}} = \frac{\lambda}{2 \sin \theta_{hkl}}$;

$$\theta_{hkl} = \sin^{-1} \left(\frac{\lambda}{2d_{hkl}} \right) = \sin^{-1} \left(\frac{\lambda}{2abc} \sqrt{h^2(bc)^2 + k^2(ac)^2 + l^2(ab)^2} \right)$$

$$2\theta_{hkl} = \sin^{-1} \left(\frac{\lambda}{d_{hkl}} \right) = \sin^{-1} \left(\frac{\lambda}{abc} \sqrt{h^2(bc)^2 + k^2(ac)^2 + l^2(ab)^2} \right)$$

Prepare a table; note that diffraction angle is $2\theta_{hkl}$.

hkl	d_{hkl} (Å)	$2\theta_{hkl}$ (°)
100	4.596	15.72

110	4.173	17.36
111	2.447	30.58
200	2.298	32.80
210	2.239	33.78
211	1.799	43.79
220	2.086	36.64
221	1.717	46.48
300	1.532	54.36
301	1.366	65.70

Q5) Designing an XRD experiment to detect phase transition (40 pts)

BaTiO₃ has a structure that is completely analogous to SrTiO₃, except that it is tetragonal. Assume that SrTiO₃ and BaTiO₃ form solid solutions, (Sr,Ba)TiO₃, but at an undetermined concentration of Ba, the solid solution takes on a tetragonal structure. Outline a powder diffraction experiment that could be used to determine the concentration of Ba at which the material transforms from cubic to tetragonal. Be sure to explain the signature that the transformation would have on the diffraction experiment.

Solution:

The cubic and tetragonal phases of the same compound will have unit cells such that $a=b=c$ in the cubic case and $a=b \neq c$ in the tetragonal case. In a cubic case, (n00) and (00n) have the same d-spacing and same diffraction angle. However, in a tetragonal case (n00) and (00n) have different d-spacings and diffraction angles, leading to formation of new diffraction peaks.

As soon as the phase transforms to tetragonal, there is appearance of new diffraction peaks, which we can compute by considering the interplanar spacing as well as Bragg's law:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$d_{hkl} = \frac{1}{\sqrt{\frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}}}$$

$$2d_{hkl}\sin\theta_{hkl} = \lambda; \theta_{hkl} = \sin^{-1}\left(\frac{\lambda}{2d_{hkl}}\right)$$